

# SEONGLAE CHO

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Senior AI Engineer with 4 years of full-stack software engineering and a first-author ICML 2026 main publication. Builds ML systems, agent infrastructure, and developer tooling, from research to production.

## EXPERIENCE

- HOLISTIC AI**, London, UK · *Senior AI Engineer* May 2025 – Present
- Built a trace-based AI agent evaluation framework with structured traces, quality criteria, and graph-based monitoring; a 300MB toxic-chat classifier beat gpt-oss-safeguard-20B on F1 (0.836 vs 0.799).
  - Built an FSM inference system reconstructing agent finite-state machines from execution traces for 95%+ behavior-coverage analysis.
  - Integrated ReBAC access control into the AI Governance platform with database-level policy enforcement.
  - Integrated agent features into an event-driven architecture (EDA) via an AWS Lambda-triggered Event Bus.
  - Promoted to Senior within one year of joining.
- KAKAO MOBILITY**, Seoul, South Korea · *Software Engineer, Digital Twin Team* Dec 2021 – Sep 2022
- Led a 3-engineer team delivering a 3D point-cloud processing service handling 10M+ point cloud frames in the autonomous-driving pipeline.
  - Built Kubernetes infrastructure with Grafana alerting and isolated namespaces.
  - Rewrote a C++ 3D-projection algorithm in Rust with Node.js N-API bindings for a 3x speedup and cross-platform deployment.
- STRYX**, Seoul, South Korea · *Software Engineer, Geospatial AI Startup* Nov 2019 – Dec 2021
- Cut build time 70% by consolidating 8 repositories into a monorepo.
  - Reduced the Docker image 90% (2GB to 180MB) via multi-stage CI builds.
- YONSEI UNIVERSITY, Data & Language Intelligence Lab**, Seoul, South Korea · *Research Intern, Prof. Dongha Lee* Mar 2023 – Jan 2024
- Interpretable-summarization research on LLMs; first-authored RTSUM (NAACL 2024 Demo).

## PROJECTS

- Vect** · local-first multi-agent coding IDE, ~80k LOC solo; orchestrates Claude Code and other coding-CLI agents in a shared workspace. Mar 2026 – Present
- MCP-Notion** · Model Context Protocol SSE server searching Notion pages and converting them to Markdown for agent interoperability. Jan 2025 – Feb 2025
- Conference Arena** ([confarena.com](https://confarena.com)) · paper-rating leaderboard web app with Elo-style ranking. Jan 2026 – Feb 2026
- TPCxAI** · scalable big-data pipelines on a Hadoop + Spark cluster in GCP for end-to-end AI workloads. Mar 2024 – May 2024
- ReSRer** · retrieval-augmented generation pipeline indexing 21M documents into a Milvus vector database. Sep 2023 – Jan 2024
- LLaMa2GPTQ** · local LLM with 4-bit GPTQ quantization (~75% memory) and RAG vector search. Jun 2023 – Jul 2023
- Texonom** · Zettelkasten knowledge system; ANN retrieval over 30,000 pages in Postgres pgvector. Nov 2021 – Jun 2023

## PUBLICATIONS

- Cho, S., Wu, Z., & Koshiyama, A. (2025). [CorrSteer](#): Generation-Time LLM Steering via Correlated Sparse Autoencoder Features. **ICML 2026 main**. +27.2% HarmBench (108 samples), +3.3% MMLU. [demo](#)
- Cho, S., Wu, Z., & Koshiyama, A. (2026). [Control Reinforcement Learning](#): Interpretable Token-Level Steering of LLMs via Sparse Autoencoder Features. ICLR 2026 Workshop on Principled Design for Trustworthy AI.
- Cho, S., Wu, Z. et al. (2026). [The Confidence Manifold](#): Geometric Structure of Correctness Representations in Language Models. ICLR 2026 Workshop on Agentic AI in the Wild. Internal probes beat output-based detection by 23–33pp AUROC across 9 LLMs.
- Cho, S. et al. (2026). PaaT (Probe as a Tool): self-regulating agents via a callable activation probe. ICML 2026 FAGen Workshop. Probe AUROC 0.93–1.00 across 11 risk heads.
- Cho, S. et al. (2026). AgentRoom: Concurrent Multi-Agent Coding in a CRDT-Backed Shared Workspace. ICML 2026 FAGen Workshop. 71% of multi-agent gain attributed to the coordination layer.
- Cho, S., Wu, Z. et al. (2026). Automata from Agent Traces: Failure and Next-Step Prediction. ICML 2026 AIWILD Workshop. 0.975 AUROC failure detection.
- Cho, S. et al. (2025). [FaithfulSAE](#): Towards Capturing Faithful Features with Sparse Autoencoders without External Datasets Dependency. ACL 2025 SRW. Outperformed external-data SAEs in 5/7 models.
- Wu, Z., Cho, S. et al. (2025). AgentGraph: Trace-to-Graph Platform for Interactive Analysis and Robustness Testing in Agentic AI Systems. AAAI 2026 Demo.
- Wu, Z., Cho, S.\* et al. (2025). [LibVulnWatch](#): A Deep Assessment Agent System and Leaderboard for Uncovering Hidden Vulnerabilities in Open-Source AI Libraries. ICML 2025 TAIG Workshop. [demo](#)
- Cho, S. et al. (2024). [RTSUM](#): Relation Triple-based Interpretable Summarization with Multi-level Saliency Visualization. NAACL 2024 System Demonstrations.

## EDUCATION

- University College London** · *MSc in AI for Sustainable Development, Distinction* Sep 2024 – Sep 2025
- Dissertation: Representation-Level Steering Improves Task Performance and Safety in LLMs. Supervisor: Prof. Philip Treleaven; Mentor: Zekun Wu.
- Yonsei University** · *BE in Computer Science* Mar 2017 – Aug 2024

## SKILLS

- AI**: Python, PyTorch, AI agents, Langfuse, LangSmith, DDP/FSDP, ONNX, PydanticAI, RAG, LangGraph, Agent SDK.
- Service**: Full-stack, TypeScript, Node.js, PostgreSQL, Redis, Rust, Faiss, Milvus, Vite, React, Next.js.
- Infra**: Kubernetes, Linux, Git, CI/CD, Docker, EDA, Apache Spark, Ansible, ETL, AWS, GCP, Hadoop, Terraform, SST.

## AWARDS

- **HERMES, 1st Place (£3,000), Holistic AI Hackathon** (Nov 2024) · cut biased, stereotypical text generation from 90% to 70% with an SAE-derived steering vector.
- **MBTIGPT, 1st Place (₩3,000,000), Yonsei GenAI Competition** (2023) · AI service with 1,000+ users, RAG over chat history with Redis and Faiss.

## SERVICE

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- Reviewer, ICML 2026 (Silver Reviewer Award) · Reviewer, ACL Rolling Review (March 2026).
- Mandatory military service completed as an Industrial Technical Personnel.